

Selection of a Suitable Thermodynamic Model for Process Simulation

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Thermodynamics of a system is the most important criteria in determining the state of the system or process. Most chemical processes rely on the design of the suitable process using appropriate thermodynamic model. An inappropriate model selection will result in convergence problems and erroneous results. Simulations are only valid when the appropriate thermodynamic model is being used. For any thermodynamic property prediction problem the following details are usually known -

- System type.
- Identity and number of components.
- The range of compositions, temperatures and pressures of the system.
- The phase(s) of interest.

Depending upon the knowledge of the system there are four categories of thermodynamic model broadly categorized as

- Equations-of-State (E-o-S)
- Activity coefficient (γ)
- Empirical
- Special system specific

However, there is no strict guideline/protocol for thermodynamic property selection. The following are some directions for selection of thermodynamic model of different systems.

Water - organic Systems

1. If no data is available UNIFAC / UNIFAC LLE is a 'reasonable' model to try. It will predict results that are consistent and it is not unreasonable to trust UNIFAC to be near the results. The more non-ideal the organic the less accurate the results. If using NRTL / UNIQUAC, for other reasons, regress the behaviour of water-organic from UNIFAC.
2. If some data is available, try to validate the UNIFAC results.
3. If some data is available and sufficiently reliable, regress binary interaction parameters (BIPs) without UNIFAC.
4. If working in ppm concentration range regress infinite dilution activity coefficients. Only use UNIFAC if it matched data and there were difficulties regressing the data into NRTL or if wanting to use UNIFAC for some other reason. If there are NRTL/UNIQUAC parameters and some data for validation, use the data to validate BIPs.

Group Contribution Method for Hydrocarbons

The two uses of the group contribution method are:

1. To determine pure component properties (T_b , T_c , P_c) for hydrocarbons, 15% accuracy is typical for straight branch, C8 and lower hydrocarbons.
2. Determining non-real binary behaviour due to size and shape of the two molecules. The UNIFAC groups and their interactions are used to calculate the liquid fugacity and to determine the size and shape of a 'part' of the molecule. Regressing BIPs for UNIFAC makes it more useful. When UNIFAC is insufficient for the components you can still use it for the vapor phase, but it fails on the liquid phase and on T-P-xy behaviour. Interaction parameters between groups dictate the interaction between groups. The overall behaviour is the sum of partial interactions. If the interactions between some of the groups for two components are not available, it means that the groups are being ignored. For example acetic acid and xylene have groups that do not have known interaction parameters, which means that part of the shape of the two is being ignored. There is no H_2O group. This means that water is considered to be an ideal circle of ideal gas law size when interacting with another component. H_2O is not a sphere, so the liquid fugacity calculation is wrong. The liquid fugacity for xylene-water and xylene-acetic acid is assumed to be 'regular solution', and the liquid fugacity for xylene-acetic acid is an incomplete result.

UNIQUAC or NRTL with vapor phase association. UNIFAC is not suitable due to lack of interaction parameters for the main groups acetic acid and xylene and there is no water subgroup.

Liquid Liquid Extraction

To select a K model that will support the calculation of two liquid phases, set the vapor/liquid/liquid/solid option on the K value screen. Models **NRTL, UNIQUAC, UNIFAC LLE** are suitable.

It is possible to regress Txy data for a pair of components but be sure to set the K flag V/L/L/S option before regressing, because different regressions are used. Txy data can also be regressed for a ternary system, such as data from a binodal.

For a discussion on alpha functions influence on the calculation of activity coefficients for the two liquid phases reference: *Smith and Van Ness, "Introduction to Chemical Engineering Thermodynamics"*.

When using an activity coefficient model to determine equilibrium between two liquid phases, the SCDS UnitOp can be used to determine separation and γ_1 , γ_2 show separation regardless of what the two phases are. The calculations for LLE on a stage are the same that would be used for VLE on a stage. There is a special UnitOp, the extractor (EXTR) which is nothing more than an SCDS unit modified to work for two liquid phases rather than one liquid phase and one vapor phase. Whenever your data is expressed as an activity coefficient, you could use the gamma regression.

The Flash UnitOp will always bring the vapor and liquid to equilibrium, which is equivalent to a Murphree Efficiency of 1.0. Binodal plots should be obtained for verification of partitioning behaviour. To do a single stage liquid - liquid extraction with Murphree efficiency applied Excel UnitOp can be used to design a special UnitOp. For such low extractions set 2 stages with efficiencies of around 0.2 to 0.3 LLE UnitOp needs a minimum of 2 stages and a single flash gives equilibrium.

Regressing BIPs for NRTL(K flag set for V/L/L/S) for LLE when using UNIFAC LLE with water present is acceptable provided the water is forming a second phase, it is always best to check with data though.

General Rules

Liquid phase water with “anything else” most times use NRTL. You can regress NRTL BIPs for the other components from UNIQUAC or UNIFAC if you don't have NRTL BIP's.

SRK is an equation of state which uses the accentric factor of a component and some 'general curve fit' parameters from “lots of lab studies” to calculate the fugacity on the basis that both molecules in the binary mixture are spheres, the only difference is how big they are. Again, water is not a sphere and it's non-ideal behavior stems from its shape as well as its size.

Peng Robinson and SRK Equations of State are good for non-water mixtures of gases and excellent for light gases and refrigerants. However they cannot handle two phase water but are good for two phase hydrocarbons. Great for non-polar, not so good for polar solutions.

You can regress a BIP for binary behaviour which is really just a mixing bip ($K_{ij}=K_i/K_j=A+B*T$) If using an EOS with water there are two methods:

1. Use NRTL and turn on either Vapor Fugacity Correction (use SRK for the vapor phase fugacity)
2. Use NRTL with Henry's component set for near supercritical light gases. Activity coefficients are good for polar solutions, if you have activity coefficient BIP's regressed.

UNIFAC is good for hydrocarbons, even polar hydrocarbons, if you have subgroups. Not good if there is a liquid phase.

Batch Distillation

For a multi-component organic-water system, with regressed vapor pressure data, the first choice would be NRTL. An equation of state is unsuitable unless the accentric factor is tuned; more details are available in the Help section.

When using the pseudo binary method to correlate the behaviour of a mixture based on the behaviour of pairs of components to each other, the situation may occur where this isn't valid (phase partitioning, ternary azeotropes) because the ternary, quaternary system behaves significantly different than pseudo binary predicts. When guessing BIPs from UNIFAC this can be a danger. When uncertain of BIPs it is not wise to make column specifications based on composition. VLE data is available from DECHEMA.

Thermodynamic Model Selection Options

⊙ IDEAL Solution	- Vapor Pressure - Equation of State
⊙ REGULAR Solution	- Equation of State
⊙ POLAR Solution	- Activity Coefficient Model
⊙ ELECTROLYTES Solution	- Electrolytes Model
⊙ POLYMER Solution	- Polymer Model

K-Value Options

⊙ IDEAL Solution — Pure component	
Vapor Pressure	Pure component

⊙ REGULAR Solution — Hydrocarbons ---- Equation of State

Soave-Redlich-Kwong	High-moderate P& T's Hydrocarbon
Chao-Seader	High-moderate P& T's Hydrocarbon
Peng-Robinson	High-moderate P& T's Hydrocarbon
API Soave	High-moderate P& T's Hydrocarbon
Grayson Streed	Moderate P& T's Hydrocarbon
Maxwell-BonnelK-Charts	Low Pressure, Heavy Hydrocarbon

⊙ NON-IDEAL (POLAR) Solution — Chemicals

UNIFAC	T=275-475K, P=0-4 atm VLE, LLE, Predictive
UNIQUAC	Highly non-ideal VLE, LLE
NRTL	Highly non-ideal, VLE, LLE
Wilson	Highly non-ideal, VLE
T. K. Wilson	Highly non-ideal, VLE, LLE
Hirnuma(HRNM)	Highly non-ideal, VLE, LLE
Wilson Salt	Non-ideal solutions with salts dissolved
Margules	Highly non-ideal, VLE, LLE (4 suffix)
Van Laar	Moderately non-ideal
Regular Solution	Moderately non-ideal (Predictive)
Modified SRK (4 parameters)	Polar compounds in regular solutions
Predictive SRK	Polar components in non-ideal solutions. better than UNIFAC at high pressures

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Amines (MEA, DEA)	Gas sweetening, Sour Water, Acid Gases and NH ₃ dissolved in H ₂ O
PPAQ	General, but electrolyte system is most common application (HCl, NH ₃ , HNO ₃)
Ideal	
Pitzer	
NRTL for electrolyte 1982	
NRTL for electrolyte 1986	

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Henry's Law	Gases dissolved in Water
TSRK	Methanol system; particularly with light gases
TEG	Dehydration of hydrocarbon using tri-ethylene-glycol
FLOR	Flory-Huggins method for polymers
UPLM	UNIFAC for polymers
ACTX	User specified activity coefficients

Polynomials	User K's
K Tables	User K's
User-Added Subroutine	User K's

Activity coefficients model

	Van Laar	Margules	UNIFAC	NRTL	UNIQUAC	Wilson
VLE	Yes	Yes	Yes	Yes	Yes	Yes
LLE	No	Yes	Yes	Yes	Yes	No
Accuracy	Good	Good	Good	Very Good	Very Good	Best
Simplicity	Simple	Simple	Complex	Complex	Complex	Complex
Calculation	Fast	Fast	Slow	Slow	Slow	Slow
BIPs	-	-	-	Need	Need	Need

There are listed below few property methods which give satisfactory convergence during simulation for various chemical applications.

Gas Processing

Application

Hydrocarbon separations
Demethanizer
C3-splitter

Cryogenic gas processing
Air separation

Gas dehydration with glycols

Acid gas absorption with
Methanol (RECTISOL)
NMP (PURISOL)

Acid gas absorption with
Water
Ammonia
Amines
Amines + methanol
(AMISOL)
Caustic
Lime
Hot carbonate

Claus process

Recommended Property Methods

PR-BM, RKS-BM, PENG-ROB, RK-SOAVE

PR-BM, RKS-BM, PENG-ROB, RK-SOAVE

PRWS, RKSWS, PRMHV2, RKSMHV2, PSRK, SR-POLAR

PRWS, RKSWS, PRMHV2, RKSMHV2, PSRK, SR-POLAR

ELECNRTL

PRWS, RKSWS, PRMHV2, RKSMHV2, PSRK, SR-POLAR

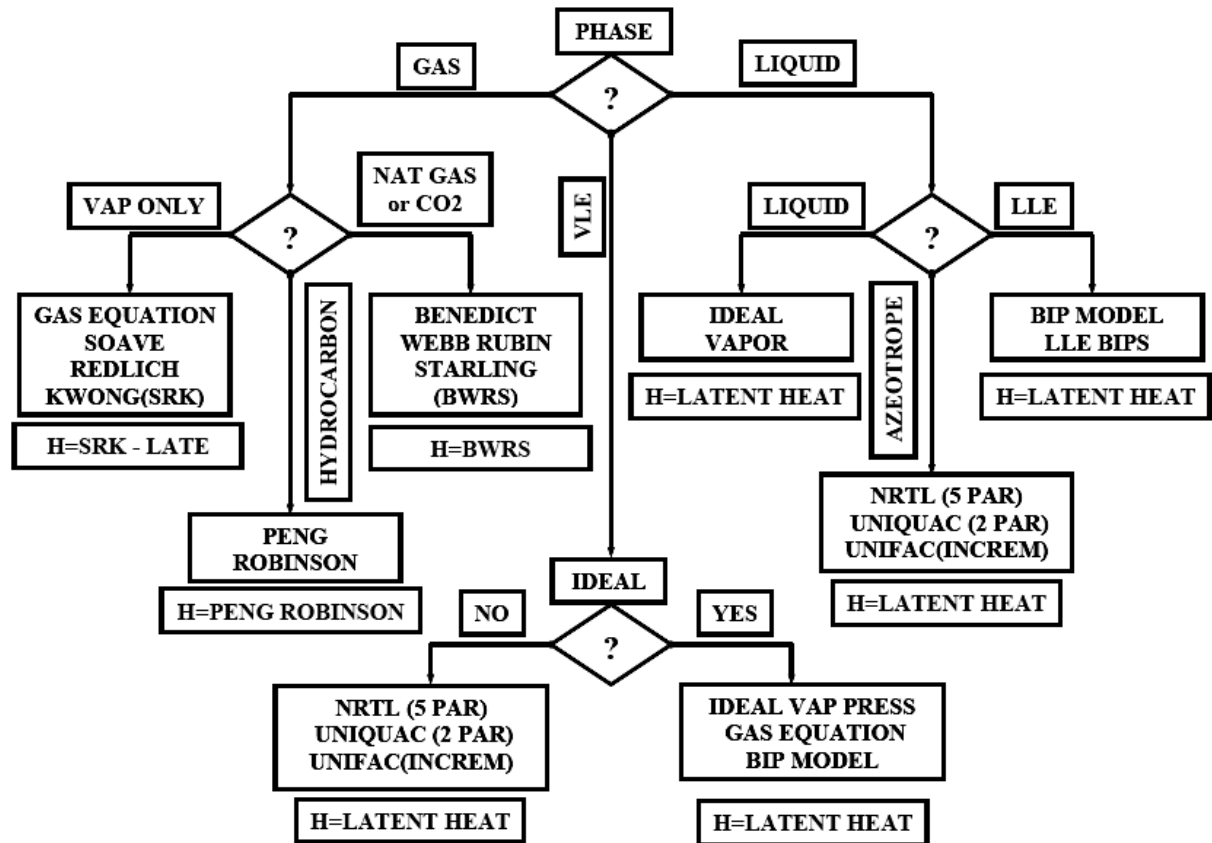
Petrochemicals

Application	Recommended Property Methods
Ethylene plant Primary fractionator	CHAO-SEA, GRAYSON
Light hydrocarbons Separation train Quench tower	PENG-ROB, RK-SOAVE
Aromatics BTX extraction	WILSON, NRTL, UNIQUAC and their variances
Substituted hydrocarbons VCM plant Acrylonitrile plant	PENG-ROB, RK-SOAVE
Ether production MTBE, ETBE, TAME	WILSON, NRTL, UNIQUAC and their variances
Ethylbenzene and styrene plants	PENG-ROB, RK-SOAVE -or- WILSON, NRTL, UNIQUAC and their variances
Terephthalic acid	WILSON, NRTL, UNIQUAC and their variances (with dimerization in acetic acid section)

Chemicals

Application	Recommended Property Methods
Azeotropic separations Alcohol separation	WILSON, NRTL, UNIQUAC and their variances
Carboxylic acids Acetic acid plant	WILS-HOC, NRTL-HOC, UNIQ-HOC
Phenol plant	WILSON, NRTL, UNIQUAC and their variances
Liquid phase reactions Esterification	WILSON, NRTL, UNIQUAC and their variances
Ammonia plant	PENG-ROB, RK-SOAVE
Fluorochemicals	WILS-HF
Inorganic Chemicals Caustic Acids Phosphoric acid Sulphuric acid Nitric acid Hydrochloric acid	ELECNRTL
Hydrofluoric acid	ENRTL-HF

K Value Model Decision Tree



Property Methods Decision Diagram

